

Are We Yet Sick of New Technologies?

The Unequal Health Implications of Workplace Digitalization *

Melanie Arntz^{1,2}, Sebastian Findeisen^{3,6}, Stephan Maurer^{4,6}, and Oliver Schlenker^{1,5}

¹Institute for Employment Research (IAB)

²Friedrich-Alexander University of Erlangen-Nuremberg

³University of Konstanz and CEPR

⁴University of Edinburgh, CEP, and UPF BSM

⁵Ifo Institute and LMU

⁶Cluster of Excellence "The Politics of Inequality"

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Abstract

This paper explores the relationship between workplace adoption of frontier technologies and workers' health outcomes, using novel and representative German linked employer-employee data. Leveraging changes in individual-level technology use between 2011 and 2019, we document that digitalization is associated with similar shifts toward more complex and service-oriented tasks across the workforce. However, only manual workers experience poorer subjective health and increased sick leave as frontier technology use rises, even after controlling for a rich set of potential confounders. In contrast, cognitive workers remain largely unaffected. Moreover, the negative technology-health association appears less pronounced in firms that provide training and support.

Keywords: health, inequality, technology, automation, tasks, capital-labor substitution

JEL: I14, J21, J23, J24, O33

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1 Introduction

Technological change has reshaped labor markets by displacing and reallocating workers across jobs (Graetz and Michaels, 2018; Bárány and Siegel, 2020; Dauth et al., 2021), inducing economic inequality by altering the wage structure (Acemoglu and Restrepo, 2022; Autor and Dorn, 2013), and transforming the tasks performed within occupations (Spitz-Oener, 2006; Acemoglu and Restrepo, 2019). While these consequences for employment and earnings have been extensively studied, much less is known about how new technologies impact workers' health.

Insights from organizational science and psychology suggest that digitalization may impose strains on workers by increasing the complexity of tasks and triggering so-called *technostress* – a stress response that arises when individuals feel unable to adapt to new digital demands (Brod, 1984; Tarafdar et al., 2015).¹ At the same time, modern technologies also have the potential to reduce physical burdens at work, for example by automating repetitive or hazardous activities. Despite this dual potential, systematic evidence on the health implications of workplace digitalization remains scarce, particularly across a broad and representative spectrum of workers, firms, and sectors.

In this paper, we address this gap by drawing on a novel and representative linked employer-employee dataset from Germany that allows us to study how the increasing use of frontier technologies at the workplace (digitalization) relates to workers' health outcomes, measured by self-reported health and sickness absence. A key strength of our data is its ability to capture heterogeneity in both the degree of technological change and the types of workers affected. Specifically, the data covers all sectors of the economy and provides rich information on individual's tasks, enabling us to distinguish between workers with predominantly cognitive, office-based task profiles and those in more manual, non-cognitive workplaces.

This distinction is essential for understanding the distributional consequences of technological change for health. Building on psychological research emphasizing that increasing task complexity can overwhelm those less accustomed to cognitively demanding work (Bakker and Demerouti, 2007; Schaufeli and Bakker, 2004), we hypothesize that digitalization is particularly stressful for low-skilled, manual workers and therefore associated with more adverse health outcomes in this

¹For example, the survey by Tarafdar et al. (2015) describes technostress as a "situation of stress experienced by the individual because of an inability to adapt to the introduction of new technology in a healthy manner". A related but distinct aspect is that exposure to technology might also increase job insecurity (see for example Dengler and Gundert (2021)). However, our evidence suggests that the health implications we find are not explained by increased job insecurity and are more likely to reflect technostress due to having to learn new technologies and having to switch to new tasks.

group.

Our empirical approach combines social security records with survey data from both firms and employees, including detailed information on the adoption and use of frontier technologies between 2011 and 2019. This allows us to construct a comprehensive measure of workplace digitalization that captures a broad range of advanced technologies, rather than focusing on any single type. We estimate first-difference models that control for major confounders, including detailed firm and job characteristics (e.g., initial tasks), as well as individual-level controls such as prior health status, earnings, technological competence, and personality traits.

This approach enables the comparison between workers in otherwise similar roles and firms, differing primarily in the pace of experienced technology adoption. While our analysis cannot establish causal effects through the absence of quasi-experimental variation, it provides representative evidence on how the diffusion of frontier technologies across the economy is linked to workers' health, thereby identifying groups that may be particularly vulnerable.

We begin by documenting how the most recent wave of digital technological progress transformed workplaces between 2011 and 2019. Our measure of workplace digitalization is most closely associated with the adoption of artificial intelligence, big data applications, and Internet of Things (IoT) technologies. These changes are accompanied by a reallocation of tasks: workers engage less in repetitive activities and more in tasks requiring social interaction, mathematical reasoning, and programming skills, while also reporting higher time and performance pressure.²

Turning to health outcomes, we find that, on average, workplace digitalization is not associated with worsening health. However, this overall result masks substantial heterogeneity: adverse health outcomes are concentrated among manual workers, who report declines in self-reported health and increases in sickness absence, while cognitive workers remain largely unaffected. The average degree of digitalization experienced by manual workers between 2011 and 2019 is associated with a decline of about 0.08 Likert scale points in self-reported health – in magnitude comparable to roughly two years of aging. This deterioration in perceived health is not only a concern for workers themselves but also matters for firms, as reduced well-being has been linked to lower productivity (Bellet et al., 2024). Consistent with this, we also observe increases in sickness absence: digitalization among manual workers corresponds to an additional 2.7 sick days in 2019, or about 20 percent relative to their baseline.

²This pattern is distinct from workplace computerization, which mainly expands the use of general ICT tools but does not substantially alter task complexity.

These patterns are not driven by differences in the types of technologies adopted or the nature of task changes across worker groups. Further, we do not observe compensating adjustments in terms of earnings or employment. As a result, workplace digitalization appears to exacerbate existing health disparities between manual and cognitive workers, echoing broader inequalities across education groups (see e.g. Meara et al. (2008); Currie (2009)) or groups of different social class (see, for example, the landmark studies by Marmot et al. (1984) and Marmot et al. (1991)).

Finally, consistent with interpreting the negative health consequences for manual workers as arising from increased technostress, we find tentative evidence that supportive firm practices – such as IT-related training and a more supportive workplace culture – attenuate these associations. This suggests that firm-level policies can play an important role in mitigating the unequal health consequences and the accompanying productivity losses arising from digitalization.

This paper relates to several strands of literature. First, it contributes to research on the consequences of technological change for workers’ health. Studies focusing on automation show that exposure to robots can reduce repetitive tasks and injuries, often benefiting low-skilled workers (Gihleb et al., 2022; Gunadi and Ryu, 2021). We expand this perspective by considering the broader spectrum of frontier technologies, many of which augment rather than substitute tasks. Our results show that while digitalization shifts work toward more complex and less automatable activities for both manual and cognitive workers, only manual workers experience deteriorations in health outcomes.

Second, our paper complements the literature linking automation risk to mental health (Blasco et al., 2024; Lordan and Stringer, 2022; Abeliansky et al., 2024). Different from these studies, which rely on task-based or industry-level exposure measures, we exploit individual-level changes in actual workplace technology use and connect them directly to workers’ well-being. Moreover, we demonstrate how declines in self-reported health are mirrored in rising sickness absence, highlighting costs for employers as well as workers.

Third, we contribute to the broader literature on how economic shocks affect health (Sullivan and Wachter, 2009; Pierce and Schott, 2020; Case and Deaton, 2021; Edin et al., 2023). Different from this literature, we do *not* focus on workers who experience prolonged periods of non-employment or sharply reduced wages, allowing us to isolate the implications of changing tasks at the workplace for health. Since we do not find significant effects of digitalization on earnings or employment, our results suggest that health consequences are more plausibly driven by evolving job environments than by deteriorating economic circumstances. This highlights a new dimension

of inequality, as digitalization appears to widen existing health gaps across education and social class (Meara et al., 2008; Currie, 2009; Marmot et al., 1984, 1991). Further, our results imply that the group of low-skilled workers is especially vulnerable as also less drastic shocks than exposure to job loss can have important implications for their welfare.

Finally, our findings connect to the expanding literature emphasizing the role of non-medical factors in shaping health inequalities (Case and Deaton, 2021; Finkelstein et al., 2021, 2024). By documenting how workplace digitalization is associated with differential health outcomes, and by showing that supportive firm practices can mitigate these disparities, we underline the importance of organizational responses in shaping the distributional consequences of technological change.

The remainder of the paper is structured as follows. Section 2 describes the data and provides descriptive statistics. Section 3 documents patterns of technology adoption and associated task changes. Section 4 outlines the empirical strategy. Section 5 presents the main results on health inequalities and explores potential mechanisms. Section 6 concludes.

2 Data and Descriptive Statistics

2.1 Data

Our analysis is based on the *Digital Transformation of Work* dataset (henceforth: DiWaBe, “Digitaler Wandel der Beschäftigung”), a novel linked employer-employee dataset that combines administrative social security records with establishment and employee surveys. The dataset was explicitly designed to capture how digital technologies diffuse into German workplaces and how they affect workers and firms.

The dataset links three components. First, an establishment survey – the *IAB-ZEW-Labor Market 4.0-Establishment Survey (BIZA)* (DOI:10.5164/IAB.FDZD.2107.de.v1) – was conducted in 2016 among a stratified random sample of 2,032 German establishments with at least one employee subject to social security contributions.³ It collected information on the use of digital technologies both in 2016 and retrospectively for 2011.

Second, a linked employee survey – the *DiWaBe Employee Survey* (DOI:10.5164/IAB.DIWABE_W01.de.en.v1) – was fielded in 2019 among the workforce of these establishments.⁴ The gross

³See Lehmer et al. (2021) for a detailed description of the BIZA survey design.

⁴For a comprehensive data report, see Müller et al. (2023).

sample consisted of 266,000 regular employees who were employed at one of the surveyed establishments on June 30, 2011 or 2016, aged between 16 and 65 in 2016, and subject to social security contributions.⁵ The net sample of the survey consists of 7,698 respondents and provides rich information on the use of workplace equipment, tasks, job requirements, training, working conditions, and health for the years 2011 and 2019. Sampling weights ensure representativeness of the German workforce in 2011 and 2016 (see Appendix B.2).

Third, both surveys are linked to administrative records from the *Integrated Employment Biographies (IEB)* of the Institute for Employment Research (IAB).⁶ These records provide precise longitudinal information on each worker’s employment spells, earnings, occupations, and industries.

The employee survey was conducted independently of employers: respondents were contacted directly by the survey company, informed about the voluntary nature of participation, and assured of strict confidentiality and anonymized use for research only. This design reduces the risk of desirability bias, especially for sensitive outcomes such as health. To facilitate recall of working conditions in 2011, the survey employed recall aids.⁷ While retrospective information may be subject to measurement error, such error is unlikely to correlate systematically with workplace digitalization and thus should, if anything, attenuate our estimates.

By combining establishment-level and employee-level survey responses with administrative records, the DiWaBe dataset provides a unique opportunity to study how changes in workplace technology adoption are associated with shifts in tasks, working conditions, and worker health, while accounting for a rich set of individual and firm-level characteristics.

2.2 Sample of Analysis

Our empirical analysis builds mainly on the DiWaBe employee survey, complemented with administrative records, and imposes several sample restrictions on the original 7,698 respondents. To ensure that exposure to digitalization stems from within-firm technological adoption rather than self-selection into technologically advanced firms, we restrict the sample to employees who

⁵The sample thus excludes civil servants, workers in minor jobs, interns, working students, and persons in vocational training.

⁶IAB Integrierte Erwerbsbiografien (IEB) V16.00.01-202012, Nürnberg 2021.

⁷An extensive pretest was conducted to examine how to best help respondents to recall 2011. As a result, respondents were first asked to recall either a personal event in 2011 and referenced to the Fukushima nuclear accident in early 2011 and the soccer World Championship in 2010 before being asked to recall their work situation. With this approach, the majority of respondents was able to recall their work situation at that time. See Appendix B.1 for details on the survey procedures and non-response.

were already employed in one of the surveyed establishments in 2011.⁸ We further confine the sample to individuals aged between 20 and 55 in 2011 in order to avoid biases from early retirement decisions. In addition, since our analysis leverages rich longitudinal information from social security records, we exclude respondents who did not consent to linking their survey responses with administrative data. Finally, individuals with missing information in any of the treatment or outcome variables are dropped. After applying these restrictions, the final sample comprises 3,235 individuals. This sample remains representative of the German workforce employed in 2011, as demonstrated in Table 17 in Appendix B.3.

2.3 Treatment Definition and Key Outcome Measures

Measuring digitalization at the workplace. Our main interest lies in the health implications of the increasing use of *advanced* digital technologies at the workplace, which we denote as *digitalization*. The DiWaBe survey allows us to distinguish digitalization from the spread of more *mature*, computer-based technologies, which we refer to as *computerization*. This distinction is crucial, since both processes involve digital tools but differ markedly in their implications for the complexity of task and potentially workers’ health.

Survey respondents were asked to report their use of workplace equipment both contemporaneously in 2019 and retrospectively for 2011. They first indicated whether and how frequently they used information and communication technologies (ICT) or machines and tools. For both types, workers then assessed the extent to which these were computer-assisted, using a five-point Likert scale.⁹

To separate advanced from mature technologies, workers who reported using computer-assisted tools were further asked whether these could also be considered “smart,” again on a five-point scale.¹⁰

We then define workplace digitalization as the extent to which an individual’s most frequently used workplace equipment became “smarter” between 2011 and 2019. Formally, the digitalization measure is constructed as a weighted average of changes in smart ICT and smart machines:

⁸We thus drop 2,625 workers who entered a surveyed establishment between 2011 and 2016.

⁹The exact wording for 2019 was: “*To what extent is the work equipment that you use computer-assisted? By ‘computer-assisted’ we mean work equipment that processes data or is equipped with a computer.*” All Likert scales are coded from 0 (low) to 4 (high).

¹⁰For those who reported no use of computer-assisted tools, we impute zero use of smart tools. Respondents were told that smart tools are “*computer-assisted tools that connect different parts of the company or production process by automatically forwarding or receiving information in order to control or optimize processes and procedures.*” Examples for tools and machines included diagnostic devices, CNC machines, robots, and 3D printers; examples for ICT included PCs, laptops, and smartphones.

$$\Delta Digi_i = w_i^{2011} \cdot \Delta ICT_i + (1 - w_i^{2011}) \cdot \Delta Machines_i, \quad (1)$$

where w_i^{2011} captures the relative importance of ICT compared to machines in 2011.¹¹ This reflects the intuition that a worker who primarily used machines in 2011 is more strongly affected by changes in machine-related technology than by ICT-related adoption. Analogously, we calculate a measure of computerization, $\Delta Comp_i$, as the weighted change in computer-assisted tools.

Relative to previous studies that focus on a specific technology (e.g., robots), these measures are more general and apply across all industries and occupations. Moreover, they allow us to distinguish between the diffusion of frontier digital technologies (digitalization) and the spread of mature, widely adopted ICT (computerization). One caveat is that the survey structure prevents a perfectly clean separation: if a worker moved from no computer-assisted tools in 2011 to a smart tool in 2019, this is mechanically recorded as both digitalization and computerization. As a result, some effects of digitalization may be absorbed into computerization. This potential misclassification, however, would bias estimates toward zero, making the differences we observe between both processes particularly striking.

Outcome variables: health measures. We examine both subjective and objective indicators of worker health. As a subjective measure, we use the change in self-assessed health between 2011 and 2019, reported on a ten-point Likert scale in each year. As an objective measure, we use the number of sickness absence days in the twelve months prior to the 2019 interview.¹² Together, these measures provide a comprehensive perspective on how digitalization is associated with workers' health, capturing both perceived well-being and more concrete indicators of illness-related absence.

2.4 Initial Workplace Heterogeneity

As discussed in the introduction, our central hypothesis is that workers in more manual and less complex jobs are particularly vulnerable to the increased task complexity and cognitive

¹¹Specifically, $w_i^{2011} = f_{ICT,i}^{2011} / (f_{ICT,i}^{2011} + f_{Machines,i}^{2011})$, where f denotes the reported frequency of use.

¹²Absence days reflect both health and institutional rules on sick pay (Maclean et al., 2020; Callison and Pesko, 2022). In Germany, the system is generous and universal, guaranteeing up to six weeks of sick leave per illness episode. During this period, the employer continues to pay 100% of wages (*Entgeltfortzahlung im Krankheitsfall*, regulated under the German Continuation of Remuneration Act). Hence, variation in sick days is likely to capture genuine differences in health rather than institutional differences.

requirements induced by digitalization. To test this, we classify workplaces into meaningful subgroups using survey information on 13 different tasks performed at the workplace in 2011.¹³

We employ a principal component analysis (PCA), a dimension-reduction method commonly used in the social sciences (see, e.g., Acemoglu et al. (2014); Sánchez De La Sierra (2020)). The PCA summarizes correlations across multiple variables into a smaller set of components that capture most of the underlying variation. This data-driven approach is particularly suitable for our setting, as we aim to identify groups of employees based on the similarity of their initial workplace tasks.

The first principal component captures 29.21% of the total variance across the 13 task measures and serves as a parsimonious yet informative summary of initial workplace heterogeneity. It loads positively on typical office-related activities such as prolonged sitting, frequent writing, and the use of software and ICT, and negatively on manual tasks involving physical exertion and machine use (see Appendix B.4 for details).

Figure 1 demonstrates that the PCA score correlates strongly with the widely used occupation-level task measures from Autor et al. (2003) (henceforth, ALM).¹⁴ Specifically, it is positively related to analytical non-routine and routine cognitive tasks, and negatively related to manual tasks. Moreover, the PCA score rises systematically with the officially defined skill requirements of occupations.¹⁵ This suggests that the PCA score provides a meaningful one-dimensional proxy for initial workplace complexity.

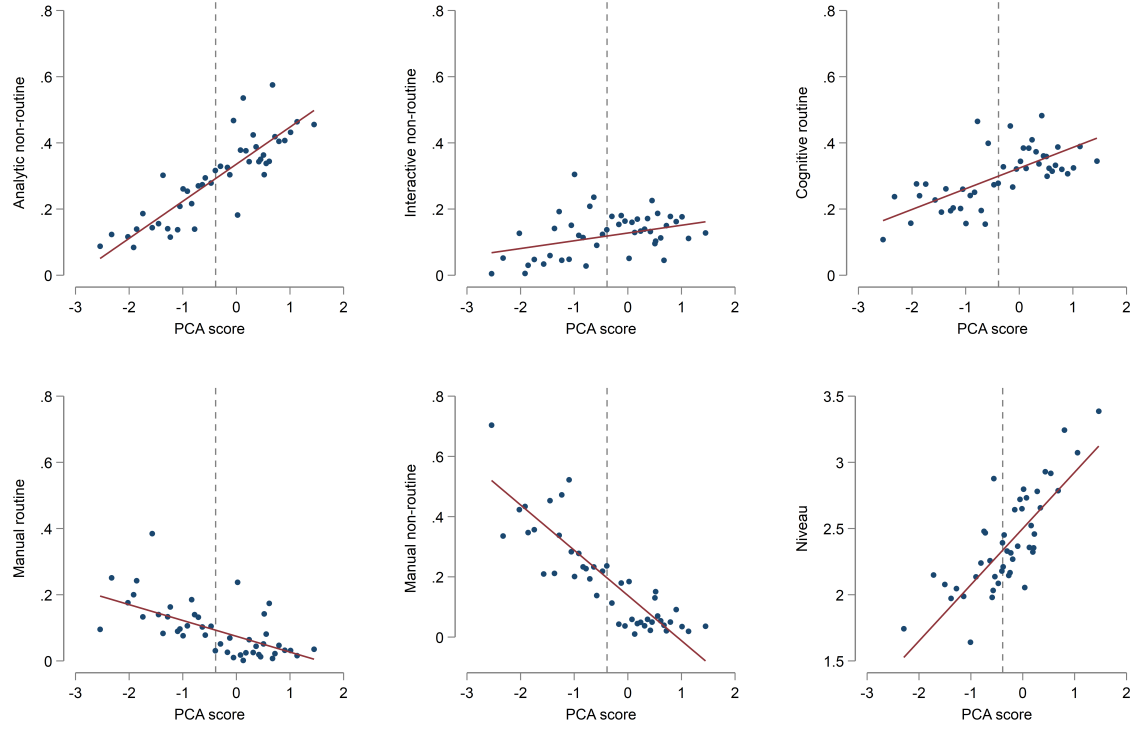
To split our sample into worker types, we again draw on the pioneering task-based classification by Autor et al. (2003). Figure 2 shows that workers in the lowest tertile of the PCA distribution are overwhelmingly in manual occupations, whereas those in the middle and upper tertiles are concentrated in cognitive or analytical occupations. We therefore classify the lowest tertile as *manual workers* and the upper two tertiles as *cognitive workers*. This classification provides a transparent and data-driven way to test whether manual workers, with less cognitively demanding initial tasks, face stronger adverse health effects from workplace digitalization.

¹³These tasks cover, among others, the frequency and degree of cognitive and physical activities, exposure to work pressure, and the scope of job autonomy. A full list and description of the variables is provided in Appendix B.4.

¹⁴For this comparison, we use occupation-level task shares obtained from Dengler and Gundert (2021).

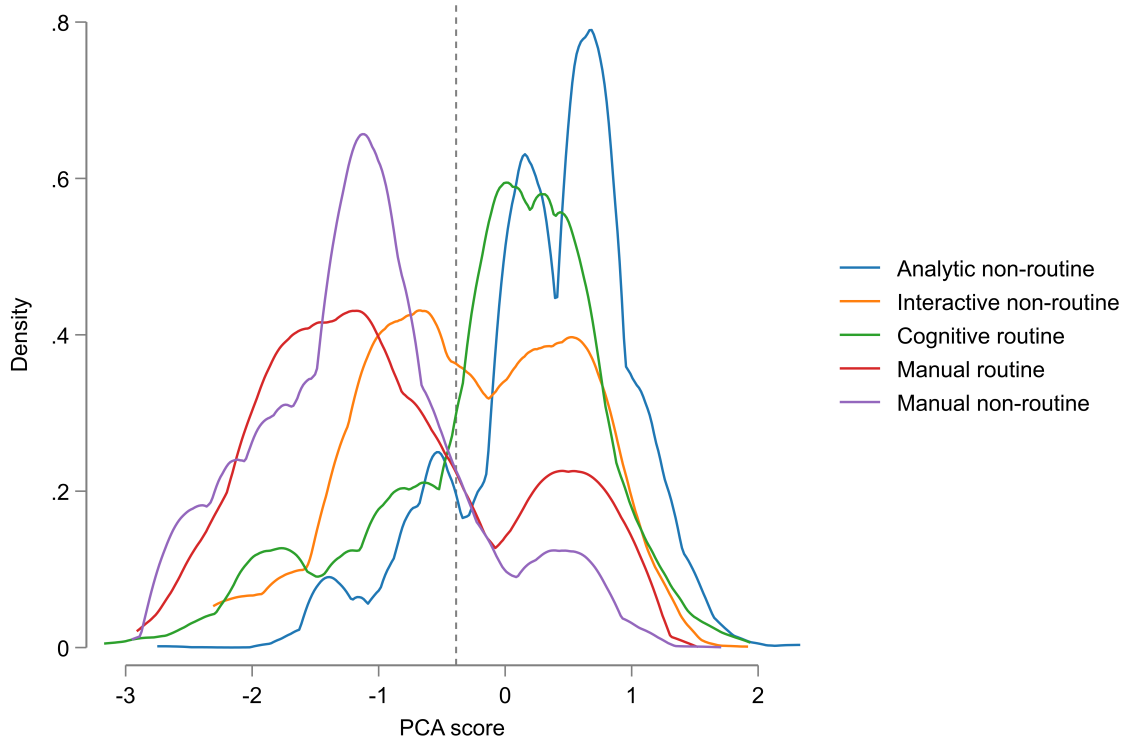
¹⁵Skill requirements are indicated by the fifth digit of the German occupation code and refer to the qualification level considered necessary by the employer, distinguishing helpers, skilled workers, specialists, and experts.

Figure 1: Comparison between ALM task categories, skill requirements, and PCA score



Notes: These graphs show the correlation between the PCA score and occupation-level task shares as well as the occupational skill requirement (denoted “niveau”). The PCA score is obtained from a principal component analysis using respondents’ information on tasks performed at their workplace in 2011 and reflects the frequency of typical office tasks (for details, see Appendix B.4). Occupation-level task shares are obtained from the BiBB/BAuA Employee Survey 2012 and divided into analytic non-routine (ANR), interactive non-routine (INR), cognitive routine (CR), manual routine (MR), and manual non-routine (MNR) tasks. These tasks are mutually exclusive and sum up to 1 for each 2-digit occupation. The occupational skill requirement is indicated by the fifth digit of individuals’ KldB occupation code in 2011. It ranges from 1 to 4, with 1 being unskilled and 4 highly complex tasks. Observations are weighted using post-stratified sampling weights, further explained in Appendix B.2.

Figure 2: Kernel-density plot of the PCA score across occupation main groups



Notes: This graph displays the kernel-density plot of the PCA score separately for each occupation main group. The PCA score is obtained from a principal component analysis using respondents' information on tasks performed at their workplace in 2011 and reflects the frequency of typical office tasks (for details, see Appendix B.4). Individuals are categorized into occupation main groups based on the most common type of tasks performed in their occupation in 2012. Occupation-level task shares are obtained from the BiBB/BAuA Employee Survey 2012 and divided into analytic non-routine (ANR), interactive non-routine (INR), cognitive routine (CR), manual routine (MR), and manual non-routine (MNR) tasks. These tasks are mutually exclusive and sum up to 1 for each 2-digit occupation. Observations are weighted using post-stratified sampling weights, further explained in Appendix B.2.

2.5 Descriptive Statistics

Table 7 in Appendix A reports descriptive statistics for individual-level controls across manual and cognitive workers. Manual workers tend to be younger, more likely male and of migrant background, and have lower formal education than cognitive workers.¹⁶ Personality traits also differ: manual workers report lower extraversion, less technological comprehension, and slightly poorer subjective health. Unsurprisingly, their average earnings are substantially lower.

Turning to firm characteristics (Table 8 in Appendix A), manual and cognitive workers are employed in establishments of similar size but with different structures. Manual workers are

¹⁶Although education is strongly correlated with worker type, our task-based classification is preferable to a purely education-based definition. In the German vocational system, a large share of workers hold a vocational degree, but their workplaces differ substantially in task requirements. Moreover, very few employees lack any degree at all. Still, we provide robustness checks using a education-based classification.

more often found in manufacturing firms with a high share of blue-collar workers and a low share of highly skilled employees. These firms typically rely more on analog tools, whereas firms employing cognitive workers are more likely to use computerized equipment.

Table 9 in Appendix A presents descriptive statistics for treatment and outcome variables. On average, workers report an increase in workplace digitalization of 0.48 points (from an initial level of 1.15 in 2011). For comparison, the baseline level of computerization in 2011 was 2.14 and rose by only 0.18 points between 2011 and 2019. Thus, while computerization advanced slowly, workplaces show a marked trend towards digitalization.

Importantly, the pace of digitalization differs sharply across worker types. Manual workers started from a much lower baseline (0.61 points in 2011) and experienced an increase of about 65%, while cognitive workers began at a higher baseline and saw an increase of about 35%. Manual workers thus face a relatively steeper rise in digitalization, in addition to increases in computerization, whereas cognitive workers' workplaces were already largely computerized in 2011.

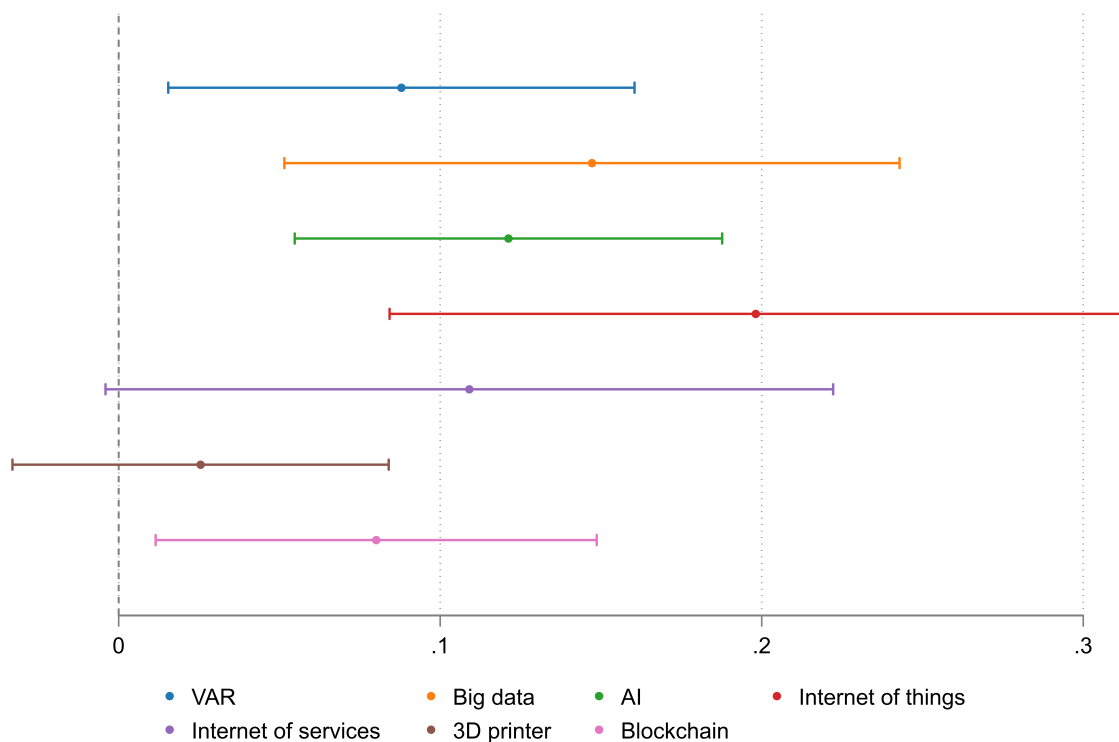
Health outcomes also differ starkly between the groups. The average worker in the sample reports 9.8 days of sick leave in 2019 and a decline of 0.36 points in subjective health (from 7.86 points in 2011), reflecting – among other factors – aging. Manual workers, however, report more than twice as many sick days as cognitive workers. In addition, they start from lower subjective health levels and experience a larger decline. This gap is consistent with well-documented health disparities across education and social class (e.g., Meara et al. (2008); Currie (2009)). Whether digitalization contributes to widening this health gap is the focus of the subsequent analysis.

3 Digitalization at the Workplace

Before turning to the health effects, it is important to establish what our measure of workplace digitalization captures. In particular, we first examine which tools and technologies digitalization is associated with. Second, we study how digitalization and computerization translate into task shifts. Finally, we assess whether these shifts differ between manual and cognitive workers. As few studies have been able to track changes in technologies and tasks at the *same workplaces* over time – rather than relying on industry or occupation aggregates or repeated cross-sections – this evidence is novel in itself.

Adoption of advanced technologies. We begin by validating our measure against the adoption of specific technologies. Respondents reported their frequency of using various digital tools in both 2011 and 2019 (measured on a five-point Likert scale). We regress the within-person change in tool use on our digitalization measure, controlling for initial workplace and firm characteristics as well as two-digit occupation fixed effects. Figure 3 shows that higher workplace digitalization is strongly associated with increased use of frontier technologies such as the Internet of Things, big data, artificial intelligence, blockchain, and virtual or augmented reality. This pattern confirms the plausibility of our measure: digitalization captures the adoption of technologies linked to the most recent wave of innovation, rather than mature technologies.

Figure 3: Digitalization and change in tools used

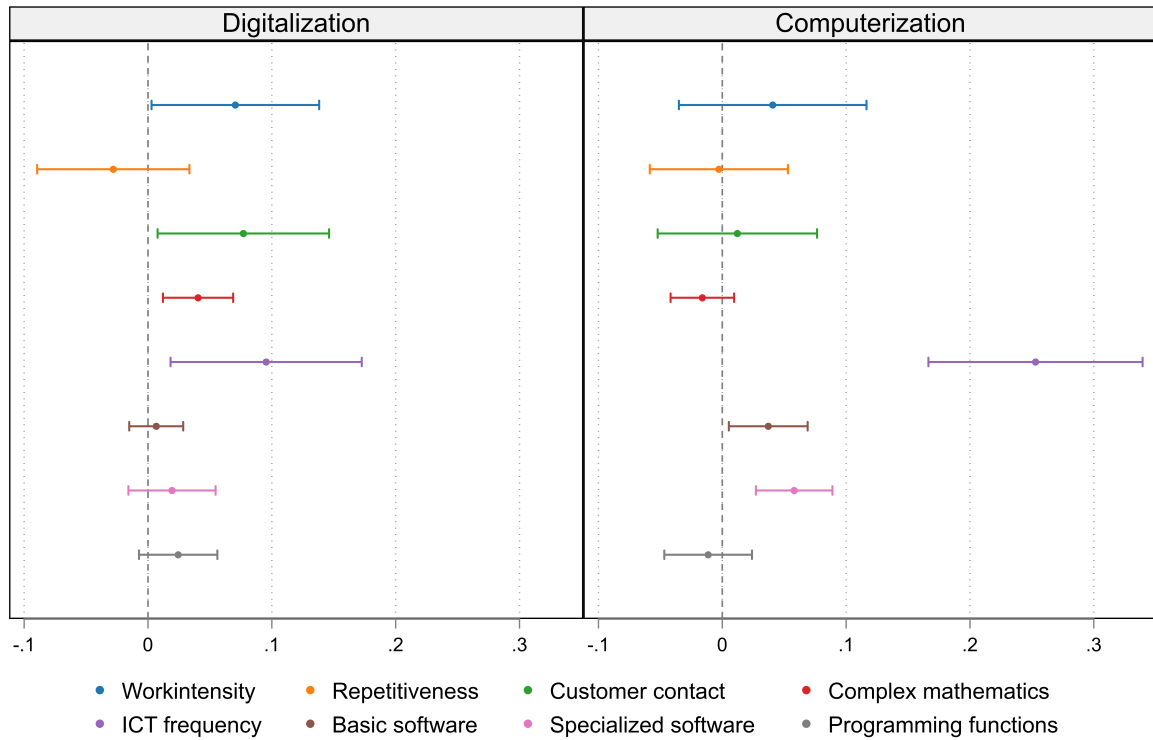


Notes: This figure displays the coefficients of separate regression of a change in the use of a specific digital tool used on digitalization. "VAR" refers to virtual and augmented reality, "AI" to artificial intelligence. The frequency of tool use is measured on a 5-point Likert scale in 2011 and 2019 and the change is calculated as the difference. Digitalization refers to a change in the use of modern digital work equipment, like smart devices. The construction of the treatment variable is explained in Section 2.3. In all regressions, we control for initial workplace and firm characteristics (as in Equation 2), as well as 2-digit occupation fixed effects. Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2. Whiskers display 95%-confidence intervals obtained from robust standard errors clustered at the firm level.

Task shifts. We next examine how digitalization reshapes the task content of jobs, using the same regression design as above. Figure 4 highlights that digitalization and computerization alter

workplaces in distinct ways, with different implications for workers' health. Computerization is linked to an increase in the use of ICT, basic software, and specialized software. By contrast, digitalization shifts jobs toward more complex tasks: more customer contact, higher demands in mathematics, more ICT use, and programming, though not standard software. The rise in advanced programming functions aligns closely with the adoption of AI and big data tools shown in Figure 3. Crucially for our analysis, digitalization is also associated with greater time and performance pressure, i.e., higher work intensity, whereas computerization has no statistically significant relationship on this dimension.

Figure 4: Digitalization and task change



Notes: This figure displays the coefficients of separate regression of a change in the performance of tasks on digitalization and computerization. In both 2019 and 2011, respondents were asked to rate (on a 5-point Likert scale) how often they (i) work under deadline or performance pressure, (ii) perform repeating identical work processes, (iii) deal with people who are not colleagues, (iv) how complex their mathematical tasks are (No calculation requirements, simple calculations, fraction/percent calculations or similar complex, area/volume calculations or similar complex, higher mathematics) and (v) how often they use ICT. Further, they were asked if they use (vi) standard software (e.g., Microsoft Office), (vii) job-specific software, or (viii) programming functions. The change is calculated as the simple difference between respondents' assessments in 2019 and 2011. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart (computer-based) devices. The construction of the treatment variables is explained in Section 2.3. In all regressions, we control for initial workplace and firm characteristics (as in Equation 2), as well as 2-digit occupation fixed effects. Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2. Whiskers display 95%-confidence intervals obtained from robust standard errors clustered at the firm level.

Treatment heterogeneity. Finally, we assess whether these patterns differ between manual and cognitive workers. To test this, we interact our measure with an indicator for cognitive workers and re-estimate the above regressions.

Appendix Figures 5 and 6 show that the changes in tools used and tasks performed are remarkably similar across worker groups. Both manual and cognitive workers report an increased use of advanced technologies and face similar task changes, particularly higher work intensity and complexity. The only notable exception is ICT frequency, which increases somewhat less among cognitive workers, likely reflecting reversion to the mean given their already high initial use.

We conclude that digitalization represents a broadly homogeneous transformation of workplaces. Differences in subsequent health effects across worker groups thus plausibly reflect differences in the ability to cope with rising complexity and intensity – not differences in the type of technologies adopted or induced task changes.¹⁷

Summary. Overall, the descriptive evidence in this section highlights three points. First, our measure of digitalization indeed captures the adoption of frontier technologies. Second, digitalization shifts jobs toward cognitively demanding, analytical, and work intensive tasks, while computerization solely reflects an increase in the frequency of software use. Third, these shifts are broadly similar for manual and cognitive workers, implying that heterogeneity in health effects stems from workers’ ability to cope with increasing complexity rather than heterogeneous exposure.

4 Estimation Strategy

The adoption of technology at the workplace is not random but reflects a combination of firm and worker decisions. Firms decide which technologies to adopt, while workers self-select into occupations, industries, and firms, potentially considering their digital skills as well as their own health. In addition, broader occupational or sectoral trends – such as trade shocks or regulatory changes – may simultaneously affect both technology adoption and health outcomes. For these reasons, identifying causal effects of technology adoption on worker health is demanding. Although we cannot make use of quasi-experimental variation, the richness of our data allows us to account for many confounders at the individual, occupation, firm, and industry level, plausibly

¹⁷We also tested whether digitalization increases perceived job insecurity. Appendix Table 15 shows no significant change in perceived risks of job loss or related concerns among manual or cognitive workers.

limiting the remaining bias.

Main Specification. Our empirical approach is based on the following regression equation for individual i in establishment j , occupation o , and industry s :

$$Y_i = \alpha + \beta \Delta Dig_i + \gamma \Delta Comp_i + \xi X_{i,j(i)} + \rho_{o(i)} + \zeta_{s(i)} + \eta F_{j(i)} + \epsilon_i, \quad (2)$$

where Y_i denotes our health outcomes. For sick leave days, we use cross-sectional data for 2019, while for subjective health, we observe the change between 2011 and 2019. The coefficient of interest is β , which captures the relationship between digitalization ΔDig_i , i.e. the workplace-level adoption of frontier digital technologies between 2011 and 2019, and health, after controlling for computerization $\Delta Comp_i$, i.e. the spread of mature computer-based technologies. We standardize both variables to have mean zero and unit variance, such that effect sizes can be interpreted in terms of standard-deviation changes. Standard errors are clustered at the firm level, as workplaces within a firm are unlikely to be independent.

Controls. To address for confounding factors, we control for a rich set of initial individual, workplace, and firm characteristics $X_{i,j(i)}$, which are either time-invariant or measured in 2011. At the individual level, controls include demographics (age, gender, education, migration background), log mean daily earnings from 2009-2011 as a proxy for productivity, Big Five personality traits, technological comprehension, and baseline subjective health as of 2011. Workplace controls include the frequency of tasks performed in 2011 (see Appendix Table 19). Firm-level controls comprise initial size and the share of blue-collar workers. A full list and detailed descriptions of all control variables are provided in Appendix Tables 7 and 8. This specification implies that identification comes from comparing workers who are similar in observable demographics, skills, and personality traits, work in comparable firms, performed similar tasks, and reported similar initial health, but were subsequently exposed to different degrees of workplace digitalization. For subjective health, we rely on first-differenced outcomes, which additionally absorbs any time-invariant, individual-specific characteristics.

Fixed Effects. Our baseline regressions include 2-digit occupation fixed effects ρ_o , which control for common trends in technology adoption and health at the occupation level. Among other factors, such as common changes in working conditions and regulations, these fixed effects

capture occupation-specific automation risks that may affect physical or mental health (see, for example, Patel et al. (2018)). In extended specifications, we add 1-digit industry fixed effects ζ_s to absorb sectoral shocks, and finally, a set of additional firm-level controls $F_{j(i)}$: workforce composition, age structure, organization, and the firm’s initial technology level (see Appendix Table 8). These variables reflect the extent to which firms are able to provide resources to help workers adjust to technological change.

Reverse Causality. A key concern to identification is reverse causality: firms may adopt digital technologies in response to the health status of their workforce, or allocate digital tools selectively to less healthy workers. Several pieces of evidence suggest this is unlikely. Conditional on occupation fixed effects, we find no systematic relationship between the adoption of digital production technologies and baseline self-assessed health in 2011. Moreover, our results are robust to including the average health change of other workers within the same firm as an additional control.¹⁸

Heterogeneity Analysis. A central hypothesis of our paper is that the health implications of digitalization are not uniform across workers, but instead depend on their initial task set. Building on the psychological literature on technostress, we expect that workers in more manual jobs are particularly vulnerable to rising digital complexity. Theories of job demands and resources emphasize that increases in cognitive demands and performance pressure can overstrain individuals who lack prior exposure or training in such environments (Brod, 1984; Tarafdar et al., 2015; Bakker and Demerouti, 2007; Schaufeli and Bakker, 2004). In contrast, workers already engaged in complex, office-based tasks may be better equipped to adapt to technological change, as they possess skills that facilitate the use of new digital tools.

To test this hypothesis, we allow for treatment heterogeneity by worker type and adapt our estimation equation as follows:

$$Y_i = \alpha + \beta_t(\Delta Dig_i \times I(t)) + \gamma \Delta Comp_i + \xi X_{i,j(i)} + \rho_{o(i)} + \zeta_{s(i)} + \eta F_{j(i)} + \epsilon_i \quad (3)$$

with $t = \{manual, cognitive\}$. $I(t)$ equals one if individual i belongs to the respective group of manual or cognitive workers, as defined in Section 2.4.¹⁹

¹⁸These results are available upon request.

¹⁹This equation is equivalent to a difference-in-differences framework, where we could display the relative effect difference between groups together with the effect for a reference group. However, as our focus is on the absolute

Channels, Compensating Differentials, and Moderating Factors. If workplace digitalization affects health, understanding the underlying mechanisms and potential compensating forces is crucial for assessing welfare and policy implications. We therefore proceed in four steps. First, we examine whether the results stem from task shifts induced by digitalization, or by the adoption of particular technologies themselves. Second, we assess whether digitalization shapes broader job perceptions, such as perceived job risks or job satisfaction. Third, we test for compensating differentials: if adverse health outcomes for manual workers are offset by higher wages or improved employment prospects, welfare consequences may be less severe. Finally, we explore moderating factors, such as training, codetermination, and support by superiors, that may mediate the association between digitalization and health.

5 The Relation between Digitalization and Health

5.1 Main Results

Table 1 presents the main results of our analysis, i.e. how workplace digitalization and computerization are associated with changes in self-reported health (columns 1-3) and with sick leave days (columns 4-6). In the first specification, shown in columns (1) and (4), we include our rich set of individual and baseline workplace characteristics together with 2-digit occupation fixed effects. In columns (2) and (5), we add 1-digit industry fixed effects to additionally absorb industry-wide shocks that may simultaneously influence the adoption of technologies and workers' health. Finally, columns (3) and (6) adds controls for the detailed structure of the firm – including the distribution of skills and age, organizational characteristics, and the firm's initial technology level (see Section 4 for details).

Across all three specifications, we find no evidence of an economically meaningful relation between digitalization and health. For subjective health, a one-standard-deviation increase in digitalization between 2011 and 2019 (equivalent to 1.13 points) is linked to a reduction of only 0.03 points on the 10-point Likert scale, or about 1.5 percent of a standard deviation. To put this number into perspective, one additional year of aging is associated with a decline of roughly 0.036 Likert points in 2019.²⁰ For sick leave days, the estimated relationship with digitalization

effect for each group, we estimate interactions of digitalization with both mutually exclusive worker types. Since we already control for each workplace characteristic in 2011, the inclusion of a worker-type fixed effect is not necessary and does not affect the results.

²⁰This coefficient stems from a bivariate regression of subjective health in 2019 on age.

is slightly positive, but again statistically insignificant and small in magnitude.

We are not surprised to find no average negative relationship as the technostress framework by Brod (1984) suggests that only workers in initially less digital and less complex jobs should experience adverse health consequences from digitalization. Since these workers account for only one third of our sample, this negative association is likely to be masked when pooling manual workers together with cognitive workers, for whom we do not expect health to deteriorate.

5.2 Heterogeneity by Worker Type

We therefore test the technostress hypothesis and estimate heterogeneous associations by worker type, using the classification of manual and cognitive workers introduced in Section 2.4. *Panel B* of Table 1 reports the results. The coefficients for cognitive workers remain small and statistically indistinguishable from zero, consistent with the notion that they do not experience adverse health consequences from digitalization.

For manual workers, however, we find sizable and statistically significant associations. A one-standard-deviation increase in digitalization between 2011 and 2019 is associated with a decline in self-reported health of about 0.22 Likert points. Taking into account that the average manual worker in our sample experienced an increase in digitalization of around 0.35 standard deviations over this period (see Table 9 in Appendix A), this corresponds to a decline in subjective health of roughly 0.08 Likert points – equivalent to about two years of aging.

Turning to sick leave, manual workers report on average 7.6 additional days absent from work for each standard-deviation increase in digitalization – thus 2.7 additional sick days for the average manual worker. Given the average of 13.5 sick days per manual worker in 2019, this corresponds to an increase of 20 percent. This increase is not only statistically meaningful but also economically relevant: using estimates from BAuA (2021), each sick day translates into direct wage costs of 123 euros and an additional 209 euros in lost productivity.

Taken together, these results support the hypothesis that the health burden of workplace digitalization is highly unevenly distributed. While cognitive workers appear resilient to technological change, manual workers face measurable declines in both subjective and objective indicators of health – in line with the technostress framework, which emphasizes that the disruptive nature of digitalization depends critically on workers’ prior exposure and ability to adapt to complex digital environments.

Table 1: Health effects

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Overall</i>						
Digitalization	-0.031 (0.066)	-0.033 (0.065)	-0.029 (0.066)	1.385 (1.101)	1.385 (1.100)	1.333 (1.085)
Computerization	-0.097 (0.062)	-0.092 (0.061)	-0.099 (0.061)	0.874 (1.071)	0.862 (1.067)	0.930 (1.093)
R-squared	0.44	0.44	0.46	0.14	0.14	0.14
<i>Panel B: By workplace type</i>						
Digitalization (Manual)	-0.237** (0.105)	-0.239** (0.106)	-0.220* (0.119)	7.680** (3.240)	7.684** (3.234)	7.567** (3.241)
Digitalization (Cognitive)	0.070 (0.076)	0.069 (0.073)	0.065 (0.071)	-1.216 (0.855)	-1.234 (0.851)	-1.229 (0.868)
Computerization	-0.079 (0.060)	-0.075 (0.059)	-0.083 (0.059)	0.253 (0.948)	0.235 (0.948)	0.303 (0.975)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.44	0.45	0.46	0.15	0.15	0.16
N	3235	3235	3235	3049	3049	3049

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Panel A shows the overall effect of digitalization and computerization. In Panel B, an interaction between an indicator for the worker type and digitalization is added to the measure of digitalization as shown in Equation 3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

5.3 Robustness

Our main results suggest that workplace digitalization is linked to deteriorating health among manual workers, whereas no such relationship emerges for cognitive workers. In this subsection, we provide a series of robustness checks to assess the credibility of these findings.

A first concern is that our results may not be specific to digitalization, but simply reflect exposure to more technology in general. As documented in Table 7, computerization remains a quantitatively important aspect of technological change at the workplace for manual workers. If the negative health-digitalization relationship is due to technostress, the increasing use of more common computer-based tools should not have any negative effect for them. While our baseline specifications always control for workplace computerization, we did not allow for heterogeneity by worker type. In Table 10 in Appendix A, we therefore interact computerization with worker type, analogous to our treatment definition for digitalization. The results remain unchanged, strengthening our interpretation that the observed relationship for manual workers is indeed specific to frontier digital technologies and their associated increase in task complexity, rather than the intensity of technology use more generally.

Further, to strengthen our interpretation that the negative relationship between digitalization and health arises from increasing complexity of tasks and tools, and not e.g. the frequency of tool usage, we control for task changes and changes in the frequency of tools used. If this increased complexity and the associated increase in time and performance pressure were the sole reason for the negative health effects of digitalization for manual workers, we would expect the latter to disappear once we control for changes in the tasks performed and the tools used at the workplace. Indeed, Appendix Table 13 shows that controlling for task shifts reduces the association with subjective health to a smaller and statistically insignificant level. The relationship with sick leave days, however, remains largely unchanged – indicating that reported illness is potentially influenced by other digitalization accompanying factors. Then, to disentangle whether health declines are due to the increasing complexity of new tools rather than their frequency of use, we further control for changes in ICT and machine use (Appendix Table 14). Results remain stable, reinforcing the interpretation that complexity rather than intensity drives the observed patterns.

Another concern is that the results may be driven by endogenous sorting of workers into firms or occupations with different technology profiles. To account for this possibility, we include indicators for whether a worker changed firm, occupation, or both between 2011 and 2019. As

shown in Appendix Table 6, the results remain essentially unchanged. This suggests that our findings are not driven by selective job mobility, but more likely reflect changes in technology exposure within continuing jobs.

Relatedly, firms could adopt digital technologies in response to workers' decreasing health, rather than technology affecting health. Two pieces of evidence mitigate this risk. First, conditional on occupation fixed effects, baseline self-assessed health in 2011 is not predictive of subsequent workplace digitalization. Second, our results remain robust when controlling for the average health change of all other workers within the same firm.²¹ Taken together, this suggests that reverse causality is unlikely to explain our findings.

Finally, we examine whether the heterogeneity by worker type is mirrored in differences by formal education. As outlined in Section 2, our baseline heterogeneity analysis is based on a PCA that classifies workers into manual and cognitive types according to the initial task composition of their workplace (see Appendix B.4 for details). Yet, as shown before, this task-based classification is strongly linked to formal education. Only one out of 10 manual workers has a college degree, compared to 40% of all cognitive workers. On the other hand, 7% of the manual workers have no vocational degree, while this share among cognitive workers is only 1%.

Table 2 therefore reports results by three education groups: unqualified workers, qualified workers with vocational training, and college graduates. The results reveal a clear gradient: workplace digitalization has sizeable negative effects on self-reported health changes and sick leave for the least educated workers while leaving more educated workers unaffected. Hence, the unequal impact of digitalization on workers' health can be seen from both a task and a skill perspective: the health burden stemming from modern technological change is concentrated on individuals with lower formal education in formerly analog workplaces. We interpret this as a first indication that education and training act as resources that help individuals cope with rising digital complexity, consistent with psychological theory.

²¹The results are available upon request.

Table 2: Health effects by education group

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Unqualified)	-0.402*	-0.419*	-0.408*	14.075**	14.092**	14.218**
	(0.226)	(0.222)	(0.218)	(5.579)	(5.574)	(5.551)
Digitalization (Qualified)	-0.089	-0.094	-0.077	0.865	0.886	0.828
	(0.090)	(0.090)	(0.092)	(1.902)	(1.928)	(1.899)
Digitalization (College)	0.092	0.095	0.087	-0.533	-0.500	-0.612
	(0.085)	(0.080)	(0.080)	(0.913)	(0.934)	(0.963)
Computerization	-0.081	-0.077	-0.085	0.451	0.423	0.521
	(0.062)	(0.061)	(0.061)	(1.125)	(1.126)	(1.141)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.44	0.45	0.47	0.15	0.15	0.16
N	3235	3235	3235	3049	3049	3049

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. For each skill level an interaction between an indicator for being in the respective education group and digitalization is added to the measure of digitalization, analogue to the interaction with worker types in Equation 3. (Un-)Qualified refers to individuals with(out) vocational degree that have not obtained a bachelor's or master's degree. College refers to individuals with at least a bachelor's degree. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

5.4 Alternative Mechanisms and Compensating Differentials.

Our findings are consistent with the technostress hypothesis: digitalization is associated with rising workplace complexity, greater cognitive demands, and higher performance pressure, all of which are negatively related to health. At the same time, these estimates could in principle also reflect other stressors that are known to impair health, such as long working hours, job insecurity, or unemployment (Goh et al., 2016). In this sense, the observed relationship may capture the combined influence of several mechanisms.

We therefore examine whether digitalization is systematically related to these alternative

stressors. Appendix Table 16 shows no evidence of changes in either actual or contracted working hours. Similarly, digitalization is not associated with shifts in workers’ perceptions of job insecurity or overall job satisfaction (Appendix Table 15). This stands in contrast to the literature on automation, which documents that exposure to robotization increased perceived job risks and thereby reduced workers’ health (Abeliansky et al., 2024). The non-labor-substituting character of digitalization is further reflected in the absence of any relationship with actual employment, as shown in Appendix Table 5. Appendix Table 5 also reports no association between digitalization and cumulative labor earnings, either on average or by worker type. This again differs from evidence on robot adoption, which has been found to exert substantial impacts on both employment and wages (Dixon et al., 2021). The absence of positive associations with earnings or employment stability implies that manual workers are not compensated for the deterioration in health they experience – suggesting a net welfare loss for affected workers.

5.5 Moderating Factors

The relationship between workplace digitalization and worker health is likely shaped by organizational structures and workplace culture, which can either amplify or cushion potential strains.²² We therefore examine whether training, the presence of works councils, and perceived support from superiors moderate the association between digitalization and health, focusing on manual workers who are most affected.

If manual workers experience health deterioration because they lack the skills required to meet rising digital demands, training may help them adapt and thereby reduce technostress. To test this, we extend Equation 2 by interacting digitalization with a dummy indicating whether the worker participated in training in the twelve months prior to the 2019 survey. Results in Table 3 show that the negative relationship between digitalization and health is concentrated among workers who did not participate in training. Among those who did, the association disappears – and in some specifications even turns slightly positive. This finding is robust to including detailed firm-level controls, implying that otherwise comparable workers in similar firms in terms of their organizational structure and level of technology face very different health outcomes from digitalization depending on whether they participate in training or not.

²²Beyond the firm level, Lauterbach et al. (2023) provide cross-country evidence that labor market institutions such as unemployment benefits may mitigate perceived technostress.

Table 3: Manual workers' health effects by individual-level training uptake

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization	-0.212 (0.138)	-0.206 (0.139)	-0.252* (0.137)	10.462*** (3.849)	11.017*** (3.622)	11.492*** (3.657)
Yes $\times$ Digitalization	0.263 (0.191)	0.234 (0.187)	0.387** (0.179)	-7.558* (4.134)	-8.052** (3.998)	-10.783** (4.338)
Computerization	-0.139 (0.108)	-0.137 (0.110)	-0.151 (0.107)	0.171 (2.610)	-0.068 (2.615)	0.152 (2.599)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.57	0.58	0.61	0.34	0.35	0.39
N	1009	1009	1009	936	936	936

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). Additionally, we restrict the sample to manual workers as only they appear to be negatively affected by technological change. Details on this classification can be found in Section 2.4. In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Throughout all specifications, we interact our measure of digitalization with an indicator of whether a manual worker participated in any training during the last 12 months prior to the interview. Robust standard errors clustered at the firm level appear in parentheses.

A limitation is that training decisions may be endogenous: workers already struggling with digitalization might self-select out of training. To address this concern, we complement the analysis with firm-reported training measures in Table 4.²³ Specifically, firms were asked in 2016 whether (i) they increasingly train employees in the use of the latest ICT technologies, (ii) they increasingly train more interdisciplinary “soft” skills relative to job-specific skills and (iii) offer employees the opportunity to gain higher qualification in the last five years (all measured on a five-point Likert scale). If digitalization impairs health through greater technological complexity, we expect a strong positive moderating role of technology-related training. On the other hand, upskilling and soft skill training could provide important resources for workers in order to deal

²³Due to sample size reductions resulting from missing information on training in the firm survey, we use the full sample in this analysis.

with increasing job demands, but we do not expect them to play a similar role in shielding workers from technostress. In line with this argumentation, adverse health outcomes are concentrated among workers in firms without extensive digital training. Although the moderating role of firm-level digital training is weaker than in the individual self-reports, the results suggest that training is not merely capturing endogenous worker selection into training.

Table 4: Health effects by firm-level training provision

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization	-0.485** (0.192)	-0.452** (0.184)	-0.448** (0.187)	7.553 (6.219)	7.496 (6.235)	7.508 (6.046)
Digitalization $\times$ ICT skills	0.131** (0.059)	0.129** (0.058)	0.144** (0.057)	-1.021 (1.570)	-1.036 (1.574)	-1.051 (1.575)
Digitalization $\times$ Soft skills	-0.027 (0.056)	-0.030 (0.054)	-0.035 (0.055)	-0.373 (0.832)	-0.359 (0.863)	-0.335 (0.826)
Digitalization $\times$ Upskilling	0.074* (0.042)	0.066 (0.041)	0.050 (0.040)	-1.232 (1.004)	-1.175 (1.046)	-1.173 (1.022)
Computerization	-0.071 (0.070)	-0.069 (0.070)	-0.077 (0.069)	0.913 (1.024)	0.856 (1.014)	0.886 (1.018)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.47	0.48	0.49	0.14	0.14	0.14
N	2617	2617	2617	2472	2472	2472

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Throughout all specifications, we interact our measure of digitalization with firms' own assessment of whether or not (i) they increasingly train employees in the use of the latest ICT technologies, (ii) they increasingly train more interdisciplinary "soft" skills relative to job-specific skills and (iii) offer employees the opportunity to gain higher qualifications on the job. All assessments are on a 5-point Likert scale, asked in 2016 and relative to 2011. Robust standard errors clustered at the firm level appear in parentheses.

Beyond training, supportive workplace institutions and culture may help workers cope with

technological change. We therefore analyze whether the presence of a works council or higher perceived support from supervisors moderates the link between digitalization and health. Results reported in Appendix Tables 12 and 11 show no systematic moderating role for works councils. By contrast, workers who feel supported by their superiors are less likely to report an increase in sickness days in response to digitalization. For subjective health, the coefficients point in the expected direction but remain statistically insignificant. These patterns suggest that a supportive firm culture may buffer some of the health strains of digitalization, though we caution against strong policy conclusions given the potential endogeneity in perceiving supervisor support.

6 Conclusion

This paper provides new evidence on the relationship between workplace digitalization and workers' health. Using a representative linked employer-employee dataset from Germany, we track the diffusion of frontier digital technologies – such as artificial intelligence, the Internet of Things, and the Internet of Services – between 2011 and 2019 and examine their implications for workers' subjective health and sickness absence. To the best of our knowledge, this is the first study to estimate health consequences of these technologies in a representative setting, moving beyond case studies that focus on specific firms and technologies. Although our results are not based on experimental or quasi-experimental variation in digitalization, our dataset allows us to control for many possible confounders while at the same time offering broader external validity than most existing studies that focus on specific firms and technologies.

We document that digitalization is associated with a shift towards more complex tasks, accompanied by higher time and performance pressure. In line with psychological theories of technostress (Brod, 1984; Tarafdar et al., 2015), we find that this shift has unequal consequences across worker groups. Manual workers, whose initial tasks are less digital and cognitively demanding, experience a significant deterioration in health, both in terms of self-assessed outcomes and sickness absence. Cognitive workers, by contrast, remain largely unaffected. A series of robustness checks confirms that these findings are not driven by sorting, reverse causality, or confounding shocks.

We further show that digitalization is not systematically related to other workplace stressors, such as increased perceived job insecurity, changes in working hours and earnings, or unemployment, which strengthens our interpretation that the observed health deterioration is

primarily linked to rising digital complexity and associated technostress. This finding leads us also to conclude that manual workers bear a disproportionate health burden, without receiving compensating differentials in terms of earnings or job stability, implying a net welfare loss for this group.

As manual workers already report worse health at baseline, this implies that digitalization exacerbates existing inequalities across education groups and social classes. This dynamic mirrors the distributional effects of automation documented for wages and employment (Acemoglu and Restrepo, 2022), but extends them to non-monetary outcomes. Since health is a key determinant of individuals' labor supply and long-term earnings potential (Currie and Madrian, 1999; García-Gómez, 2011), our findings suggest that digitalization may reinforce economic disadvantages between these groups.

At the same time, our results indicate that organizational factors attenuate these relationships. Both individual participation in training and firm-level provision of digital skill development are associated with a reduced decline in health, and perceived support by supervisors similarly mitigates adverse outcomes. These patterns suggest that the health burden of digitalization is not inevitable and underscores the importance of workplace-level interventions in shaping the distributional consequences of technological change. However, further research is needed to better understand how re-skilling and task redesign can effectively prevent technology shocks from translating into health shocks.

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A Appendix: Additional Tables and Figures

Table 5: Economic effects

	Log days employed			Log labor earnings		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Overall</i>						
Digitalization	0.000	0.002	0.001	0.000	0.004	0.004
	(0.008)	(0.008)	(0.008)	(0.013)	(0.013)	(0.013)
Computerization	0.015**	0.015**	0.016**	0.040***	0.039***	0.043***
	(0.007)	(0.007)	(0.007)	(0.013)	(0.013)	(0.013)
R-squared	0.27	0.27	0.28	0.65	0.66	0.66
<i>Panel B: By workplace type</i>						
Digitalization (Manual)	0.013	0.014	0.015	0.015	0.015	0.019
	(0.011)	(0.011)	(0.012)	(0.020)	(0.020)	(0.021)
Digitalization (Cognitive)	-0.005	-0.004	-0.004	-0.007	-0.002	-0.003
	(0.010)	(0.010)	(0.010)	(0.016)	(0.016)	(0.015)
Computerization	0.014*	0.013*	0.015**	0.039***	0.038***	0.042***
	(0.007)	(0.007)	(0.007)	(0.013)	(0.013)	(0.013)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.27	0.28	0.28	0.65	0.66	0.67
N	3167	3167	3167	3167	3167	3167

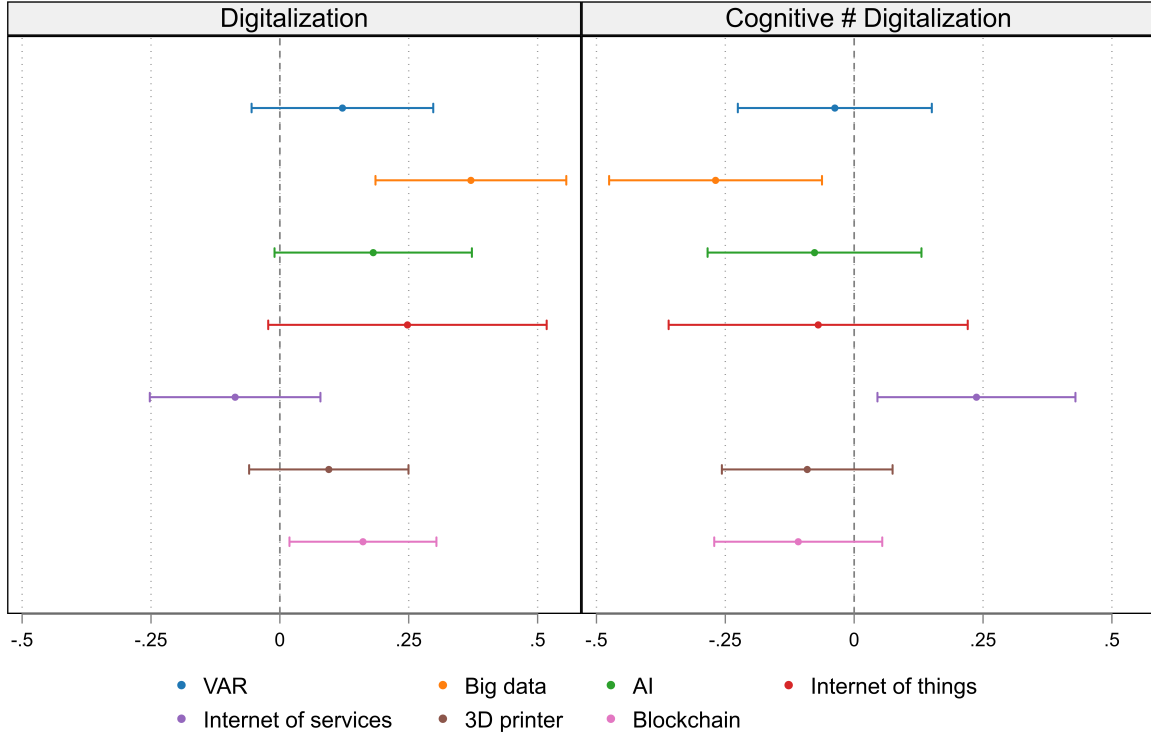
Notes: This table reports the effect of technological change on the log total numbers of days employed (columns (1) - (3)) and the log cumulative earnings between 2012 and 2019 (columns (4) - (6)). Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Panel A shows the overall effect of digitalization and computerization. In Panel B, an interaction between an indicator for the worker type and digitalization is added to the measure of digitalization as shown in Equation 3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

Table 6: Health effects after controlling for occupation and firm switching

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Manual)	-0.238** (0.103)	-0.242** (0.103)	-0.222* (0.116)	7.689** (3.190)	7.684** (3.187)	7.572** (3.201)
Digitalization (Cognitive)	0.064 (0.075)	0.063 (0.072)	0.059 (0.070)	-1.327 (0.871)	-1.326 (0.867)	-1.319 (0.883)
Computerization	-0.075 (0.060)	-0.071 (0.059)	-0.079 (0.059)	0.326 (0.960)	0.300 (0.960)	0.346 (0.982)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.45	0.45	0.47	0.16	0.16	0.16
N	3235	3235	3235	3049	3049	3049

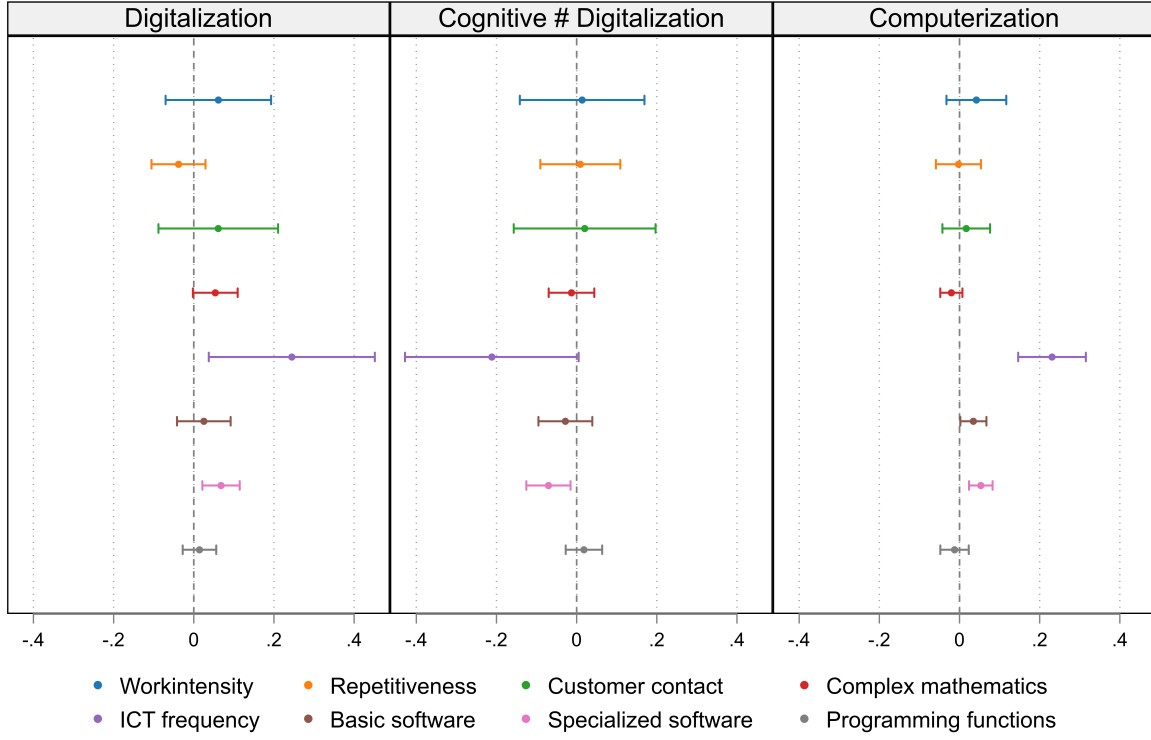
Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. For each worker type an interaction between an indicator for the type and digitalization is added to the measure of digitalization as shown in Equation 3. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Additionally, we include a dummy for whether respondents switched firms, occupations, or both. Robust standard errors clustered at the firm level appear in parentheses.

Figure 5: Digitalization and change in tools used by PCA score



Notes: This figure displays the coefficients of separate regression of a change in the use of a specific digital tool used on digitalization, overall and interacted with a dummy variable that equals one if a person belongs to the group of cognitive workers. "VAR" refers to virtual and augmented reality, "AI" to artificial intelligence. The frequency of tool use is measured on a 5-point Likert scale in 2011 and 2019 and the change is calculated as the difference. Digitalization refers to a change in the use of modern digital work equipment, like smart devices. The construction of the treatment variable is explained in Section 2.3. Cognitive workers are defined as those in the upper two tertiles of the PCA score distribution as described in 2.4. In all regressions, we control for initial workplace and firm characteristics (as in Equation 2), as well as 2-digit occupation fixed effects. Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2. Whiskers display 95%-confidence intervals obtained from robust standard errors clustered at the firm level.

Figure 6: Digitalization and task change by PCA score



Notes: This figure displays the coefficients of separate regression of a change in the performance of tasks on digitalization, overall and interacted with a dummy variable that equals one if a person belongs to the group of cognitive workers. In both 2019 and 2011, respondents were asked to rate (on a 5-point Likert scale) how often they (i) work under deadline or performance pressure, (ii) perform repeating identical work processes, (iii) deal with people who are not colleagues, (iv) how complex their mathematical tasks are (No calculation requirements, simple calculations, fraction/percent calculations or similar complex, area/volume calculations or similar complex, higher mathematics) and (v) how often they use ICT. Further they were asked if they use (vi) standard software (e.g. Microsoft Office), (vii) job-specific software, or (viii) programming functions. The change is calculated as the simple difference between respondents' assessment in 2019 and 2011. Digitalization refers to a change in the use of modern digital work equipment, like smart devices. The construction of the treatment variable is explained in Section 2.3. Cognitive workers are defined as those in the upper two tertiles of the PCA score distribution as described in 2.4. In all regressions, we control for initial workplace and firm characteristics (as in Equation 2), as well as 2-digit occupation fixed effects. Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2. Whiskers display 95%-confidence intervals obtained from robust standard errors clustered at the firm level.

Table 7: Summary statistics: Individual-level controls

	Manual	Cognitive	Total
Age (in 2011)	38.80 (9.89)	41.15 (8.67)	40.08 (9.32)
Female	0.32 (0.47)	0.51 (0.50)	0.42 (0.49)
Migrant	0.10 (0.30)	0.06 (0.23)	0.08 (0.27)
No vocational degree	0.07 (0.26)	0.01 (0.11)	0.04 (0.19)
Vocational degree	0.83 (0.38)	0.59 (0.49)	0.70 (0.46)
College degree	0.10 (0.30)	0.40 (0.49)	0.26 (0.44)
Extraversion	2.79 (0.93)	3.10 (0.91)	2.96 (0.93)
Agreeableness	2.05 (0.93)	2.16 (0.69)	2.11 (0.81)
Conscientiousness	2.20 (0.61)	2.24 (0.51)	2.23 (0.56)
Emotional Stability (Low Neuroticism)	3.15 (0.83)	3.04 (0.78)	3.09 (0.80)
Openness to Experiences	2.99 (0.89)	3.07 (0.93)	3.03 (0.91)
Technocomprehension 2011	2.45 (1.45)	2.71 (1.01)	2.59 (1.24)
PCA score	-1.27 (0.59)	0.40 (0.43)	-0.37 (0.98)
Subjective health 2011	7.80 (1.78)	7.92 (1.72)	7.86 (1.75)
Log mean daily earnings 2009-2011	4.22 (0.44)	4.58 (0.52)	4.42 (0.52)

Notes: This table contains the mean and standard deviations (in parentheses) of all individual-level variables, which are used as controls in the baseline version of Equation 2 and 3 separately displayed for the group of manual workers, cognitive workers, and over the whole sample. Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2. The Big Five personality traits (extraversion, agreeableness, conscientiousness, emotional stability, and openness to experiences) are calculated as simple mean based on the 10-item inventory and 5-point Likert scales (from 0 to 4). The PCA score captures the degree to which workplaces contain typical office tasks in 2011 and its calculation is described in Appendix B.4.

Table 8: Summary statistics: Firm-level controls

	Manual	Cognitive	Total
<i>Baseline firm controls</i>			
Firm size <50	0.46 (0.50)	0.44 (0.50)	0.45 (0.50)
Firm size 50-199	0.23 (0.42)	0.24 (0.42)	0.24 (0.42)
Firm size >=200	0.31 (0.46)	0.33 (0.47)	0.32 (0.47)
Share of blue-collar workers 2011	0.38 (0.34)	0.17 (0.25)	0.27 (0.31)
<i>Additional firm controls</i>			
Manufacturing establishment	0.33 (0.47)	0.21 (0.41)	0.27 (0.44)
Public entity	0.08 (0.27)	0.12 (0.33)	0.10 (0.30)
Trainee program	0.64 (0.48)	0.78 (0.41)	0.72 (0.45)
Age structure (share above 55)	0.11 (0.09)	0.13 (0.11)	0.12 (0.10)
Skill structure (share high-qualified)	0.11 (0.14)	0.27 (0.24)	0.20 (0.22)
Firm-level share 2011: Analogue	54.29 (26.28)	42.14 (22.94)	47.60 (25.23)
Firm-level share 2011: Computerized	33.89 (24.76)	48.79 (24.28)	42.06 (25.59)
Firm-level share 2011: Digitalized	5.73 (14.47)	6.41 (12.68)	6.10 (13.52)

Notes: This table contains the mean and standard deviations (in parentheses) of all firm-level variables, which are used as controls in either the baseline or an advanced version of Equation 2 and 3 separately displayed for the group of manual workers, cognitive workers, and over the whole sample. Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2. Shares of technologies are based on the assessment of firms to which degree their office tools/machines are analogue/manually operated, IT-based/indirectly operated, or IT-integrated/operate autonomously (from 0 to 100). Similar to the worker level, these shares are weighted between machines and ICT by the share of blue-collar workers in 2011.

Table 9: Summary statistics: Outcome and treatment variables

	Manual	Cognitive	Total
<i>Outcome variables</i>			
Subjective health change	-0.46 (2.11)	-0.29 (1.83)	-0.36 (1.96)
Days sick at work 2019	13.52 (29.63)	6.83 (20.94)	9.84 (25.43)
Log total days employed 2012-2019	7.81 (0.42)	7.88 (0.26)	7.85 (0.35)
Log total labor earnings 2012-2019	12.20 (0.65)	12.61 (0.57)	12.43 (0.64)
<i>Technological change</i>			
Computerization	0.36 (1.15)	0.04 (1.01)	0.18 (1.09)
Digitalization	0.40 (1.13)	0.56 (1.12)	0.48 (1.13)
<i>Baseline technology</i>			
Computerization 2011	1.16 (1.20)	2.96 (0.92)	2.14 (1.39)
Digitalization 2011	0.61 (0.98)	1.60 (1.29)	1.15 (1.26)
Time share 2011: Computerized	14.90 (20.46)	40.56 (28.32)	28.94 (28.13)
Time share 2011: Digitalized	8.10 (16.55)	25.72 (29.91)	17.74 (26.27)

Notes: This table contains the mean and standard deviations (in parentheses) of all outcome and treatment variables, which are used in the empirical analysis, separately displayed for the group of manual workers, cognitive workers, and over the whole sample. Additional variable on the use of technology in 2011, which are not used in the analysis, are displayed at the bottom. As for technological change, digitalization/computerization is calculated as the weighted mean of the degree to which ICT and machines are smart/computer-based. Note that while we use a standardized measure of digitalization and computerization in the analysis, with a mean of 0 and a standard deviation of 1, here we display unstandardized values. Time shares refer to workers' assessment of their "working time spent using non-computerized, computerized or intelligently connected work equipment" (0-100). Observations are weighted using trimmed post-stratified sampling weights, further explained in Appendix B.2.

Table 10: Controlling for group-specific computerization

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Manual)	-0.195 (0.120)	-0.201 (0.122)	-0.176 (0.139)	7.797** (3.550)	7.838** (3.538)	7.652** (3.538)
Digitalization (Cognitive)	0.062 (0.075)	0.062 (0.073)	0.057 (0.070)	-1.269 (0.828)	-1.294 (0.820)	-1.272 (0.836)
Computerization (Manual)	-0.151 (0.116)	-0.142 (0.117)	-0.158 (0.125)	0.045 (2.399)	-0.036 (2.438)	0.154 (2.435)
Computerization (Cognitive)	-0.053 (0.067)	-0.050 (0.065)	-0.054 (0.063)	0.429 (0.720)	0.441 (0.738)	0.448 (0.767)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.44	0.45	0.46	0.15	0.15	0.16
N	3235	3235	3235	3049	3049	3049

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. For each worker type an interaction between an indicator for the type and digitalization is added to the measure of digitalization as shown in Equation 3. In this table, this is done analogously for computerization. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

Table 11: Effect moderation: Support by superior

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization	-0.330*	-0.348*	-0.262	13.072***	13.726***	12.796***
	(0.183)	(0.189)	(0.195)	(3.318)	(3.339)	(3.248)
Yes $\times$ Digitalization	0.223	0.227	0.105	-7.499*	-7.794*	-7.674*
	(0.202)	(0.203)	(0.208)	(4.459)	(4.224)	(4.337)
Computerization	-0.076	-0.067	-0.111	-0.311	-0.675	-0.108
	(0.106)	(0.108)	(0.107)	(2.475)	(2.560)	(2.570)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.60	0.60	0.64	0.34	0.35	0.39
N	993	993	993	921	921	921

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Digitalization is further interacted with an indicator that equals one if an individual reported feeling supported by their superior in 2019. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In this table, we further restrict our sample to workers classified as manual workers based on the 2011 workplace. Details on this classification can be found in Section 2.4. In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

Table 12: Effect moderation: Firm has a works council

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization	-0.013 (0.222)	0.008 (0.233)	0.039 (0.224)	16.118* (9.598)	16.698* (10.041)	15.894 (10.028)
Yes $\times$ Digitalization	-0.148 (0.210)	-0.178 (0.218)	-0.214 (0.213)	-9.212 (9.597)	-9.539 (10.130)	-9.512 (10.207)
Computerization	-0.150 (0.112)	-0.149 (0.112)	-0.156 (0.111)	0.260 (2.739)	0.148 (2.812)	0.575 (2.698)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.58	0.58	0.61	0.34	0.36	0.39
N	992	992	992	919	919	919

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Digitalization is further interacted with an indicator that equals one if a firm has a works council established. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In this table, we further restrict our sample to workers classified as manual workers based on the 2011 workplace. Details on this classification can be found in Section 2.4. In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

Table 13: Health effects conditional on task changes

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Manual)	-0.152 (0.127)	-0.162 (0.122)	-0.144 (0.130)	7.878** (3.325)	8.169** (3.334)	8.153** (3.299)
Digitalization (Cognitive)	0.046 (0.079)	0.058 (0.076)	0.064 (0.076)	-0.779 (1.116)	-0.731 (1.109)	-0.598 (1.145)
Computerization	-0.117* (0.068)	-0.115* (0.066)	-0.134** (0.066)	0.989 (1.103)	0.869 (1.077)	1.075 (1.117)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.48	0.49	0.50	0.25	0.25	0.26
N	2659	2659	2659	2520	2520	2520

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. For each worker type an interaction between an indicator for the type and digitalization is added to the measure of digitalization as shown in Equation 3. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Additionally, we control for task changes between 2019 and 2011 (for a detailed overview, see the notes of Figure 4). Robust standard errors clustered at the firm level appear in parentheses.

Table 14: Health effects conditional on change in frequency of ICT and machine use

	Change of subjective health			Sickness absence days		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Manual)	-0.223** (0.102)	-0.224** (0.103)	-0.204* (0.116)	7.774** (3.333)	7.762** (3.318)	7.677** (3.326)
Digitalization (Cognitive)	0.073 (0.076)	0.072 (0.073)	0.068 (0.071)	-1.152 (0.842)	-1.165 (0.835)	-1.155 (0.849)
Computerization	-0.071 (0.061)	-0.064 (0.060)	-0.072 (0.061)	0.214 (1.069)	0.187 (1.073)	0.267 (1.107)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.44	0.45	0.46	0.15	0.15	0.16
N	3235	3235	3235	3049	3049	3049

Notes: This table reports the effect of technological change on a change in individuals' subjective health between 2019 and 2011 (columns (1) - (3)) and the number of sickness absence days in 2019 (columns (4) - (6)). Subjective health is measured in each year on a 10-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. For each worker type an interaction between an indicator for the type and digitalization is added to the measure of digitalization as shown in Equation 3. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Additionally, we control for the change in the frequency of ICT and machine use (asked in both 2019 and 2011 on a 5-point Likert scale) Robust standard errors clustered at the firm level appear in parentheses.

Table 15: Worries about job

	Job satisfaction		Job in danger		Job sorrow	
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Manual)	0.087 (0.108)	0.093 (0.122)	-0.023 (0.021)	-0.022 (0.022)	-0.013 (0.027)	-0.019 (0.029)
Digitalization (Cognitive)	0.069 (0.077)	0.045 (0.071)	-0.006 (0.013)	-0.003 (0.013)	-0.015 (0.015)	-0.012 (0.014)
Computerization	-0.163*** (0.061)	-0.156*** (0.060)	-0.009 (0.009)	-0.009 (0.010)	0.013 (0.012)	0.012 (0.012)
Further controls	Occupation	Industry	Occupation	Industry	Occupation	Industry
R-squared	0.21	0.23	0.19	0.24	0.24	0.27
N	3234	3234	3215	3215	2993	2993

Notes: This table reports the effect of technological change on job satisfaction (columns (1) - (2)), workers' assessment of the likelihood their job being at risk (columns (3) - (4)), and the degree to which workers have actual sorrows about losing their job (columns (5) - (6)). Job satisfaction is measured in 2019 on a 10-point Likert scale. The other variables are measured on a 5-point Likert scale. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. For each worker type an interaction between an indicator for the type and digitalization is added to the measure of digitalization as shown in Equation 3. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1), (3), and (5), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2), (4), and (6) we add one-digit industry fixed effects. Robust standard errors clustered at the firm level appear in parentheses.

Table 16: Working time

	Change of real workingtime			Change of contracted workingtime		
	(1)	(2)	(3)	(4)	(5)	(6)
Digitalization (Manual)	0.023 (0.488)	0.084 (0.481)	0.022 (0.483)	0.080 (0.294)	0.096 (0.295)	0.053 (0.294)
Digitalization (Cognitive)	0.291 (0.322)	0.458 (0.317)	0.529* (0.311)	0.688** (0.303)	0.738** (0.303)	0.754** (0.293)
Computerization	0.367 (0.258)	0.311 (0.266)	0.223 (0.251)	0.069 (0.190)	0.058 (0.192)	0.033 (0.187)
Further controls	Occupation	Industry	Firm	Occupation	Industry	Firm
R-squared	0.18	0.22	0.24	0.18	0.19	0.20
N	3133	3133	3133	3195	3195	3195

Notes: This table reports the effect of technological change on the change in hours actually worked per week (columns (1) - (3)) and contracted weekly working hours (columns (4) - (6)) based on the difference between workers' self-assessment in 2019 and 2011. Digitalization (computerization) refers to a change in the use of modern (mature) digital work equipment, like smart devices (computers). The construction of the treatment variables is explained in Section 2.3. Based on initial workplace in 2011 and using a PCA in order to reduce dimensions, respondents were divided into two main worker types: manual and cognitive workers. For each worker type an interaction between an indicator for the type and digitalization is added to the measure of digitalization as shown in Equation 3. Details on this classification can be found in Section 2.4. The sample consists of respondents from the DiWaBe employee survey conducted in 2019, linked to IAB administrative records and a firm survey. It consists of individuals in regular employment in one of the surveyed firms in 2011, aged 20-55 in 2011, giving consent to linking with administrative records, and having non-missing responses in treatment and outcome variables. To ensure that results are representative for the German workforce, we use post-stratified trimmed survey weights (see Appendix B.2 for details). In columns (1) and (4), we control for individual- and baseline firm-level controls as well as 2-digit occupation fixed effects. In columns (2) and (5) we add one-digit industry fixed effects. In columns (3) and (6) we further add controls reflecting the firm structure (see Section 4 for details). Robust standard errors clustered at the firm level appear in parentheses.

B Appendix: Data

B.1 DiWaBe: Survey participation and non-response

The CATI interviews for the DiWaBe employee survey were conducted by the survey provider *Umfragezentrum Bonn GmbH* between July and November 2019. All workers drawn from the sample population received a written invitation for the survey before being called. 16.4 percent of all invited workers participated in the survey, resulting in 7,900 interviews. On average, an interview lasted 37.5 minutes. Non-participation was typically due to general refusal of interviews, especially on the phone, as well as due to time restrictions. However, the topic of the survey was largely irrelevant to non-participation, which gives confidence that non-response was not influenced by awareness of or interest in the topic. Still, non-response was systematically lower among workers with a tertiary degree, and we take account of this in the sampling weights, see Appendix B.2.

B.2 DiWaBe: Sampling and Weights

Our analysis is based on individuals who took part in the DiWaBe employee survey. As these respondents were in turn selected on the basis of a firm survey, weighting is essential to ensure the representativeness of the final sample and by that ensure that our descriptive statistics and treatment effects are also representative. The details of sampling and weighting are described in the following.

Firstly, the firms were selected from the potential group of participants in the "IAB-ZEW Arbeitswelt 4.0" firm survey, stratified into 40 cells by region, firm size class and sector. From these 2,032 firms, all employees subject to social insurance contributions aged 16 to 65 who had an employment spell in 2011, 2016 or both were drawn. The gross sample of 266,000 employees was further divided into three groups: Stayers, Leavers and Entrants. For each group, the employees were in turn divided into a total of 81 stratification cells based on the following stratification characteristics: company size class (small/medium/large), education (low/medium/high) and age (young/medium/old).

As the drawing design of the gross sample without weighting is not representative of all dependent employees in Germany, design weights were created. The basis for calculating the design weights of a respondent is the inclusion probability of their firm within the 40 strata of the firm survey. The inclusion probability of a firm from a specific stratum is calculated as the

ratio of the number of surveyed firms to the total number of firms in the population in the same stratum. As the final net sample for the employee survey is again a stratified simple random sample from the firms, which were again drawn stratified, both draws must be taken into account simultaneously. The probability of inclusion was estimated for each employee from a specific stratum of the employee sample in a specific establishment on the basis of the total number of employees and the number of employees in the respective stratum. The design weights of an employee from a particular stratum were calculated as the inverse of the estimated inclusion probability of this stratum. This value therefore reflects the probability of an employee from a specific stratum and a specific establishment being included in the sample.

In order to adjust the distribution of the stratum variables of the employee sample to the distribution in the population of all employees in Germany and to compensate for the non-response that occurred, an additional post-stratification was carried out. For this purpose, the number of employees in each stratum of the population was divided by the number of employees in the same stratum of the net sample, i.e. the persons ultimately surveyed. These post-stratified design weights are used for all descriptive statistics. For the main analysis, however, we use a trimmed version. Trimming the weights at the 99th percentile makes the analysis less susceptible to outliers and represents a compromise between outlier sensitivity and bias due to weight adjustment. Details on the stratified sampling of respondents, the calculation of the weights and their effect can be found in Chapter 2.5 in Arntz et al. (2020).

B.3 DiWaBe: Representativeness

Table 17 compares major characteristics of the unweighted and weighted sample of analysis with the corresponding population characteristics. The population refers to the entire German workforce employed in 2011 for whom the sample restrictions discussed in Section 2 apply. As can be seen, especially the weighted sample matches the overall population quite well, but we have a slight over-representation of college graduates and an under-representation of older workers. The under-sampling of older workers is due to the fact that for our analysis, we include only workers aged between 20 and 55 in 2011 to ensure that the health effects are not contaminated by (early) retirement.

Table 17: Representativeness of the final sample

	Population	Weighted	Unweighted
<i>Education</i>			
No vocational degree	0.09	0.04	0.02
Vocational degree	0.74	0.70	0.48
College degree	0.17	0.26	0.49
<i>Age</i>			
Age <35	0.29	0.32	0.35
Age 35-49	0.45	0.50	0.42
Age ≥50	0.26	0.17	0.23
<i>Firm size</i>			
Firm size <50	0.44	0.45	0.16
Firm size 50-199	0.24	0.23	0.34
Firm size ≥200	0.32	0.32	0.50
<i>Location</i>			
East Germany	0.19	0.19	0.47
<i>Sector</i>			
Secondary sector, non knowledge-intensive	0.20	0.29	0.16
Secondary sector, knowledge-intensive	0.10	0.09	0.22
Tertiary sector, non knowledge-intensive	0.48	0.39	0.17
Tertiary sector, knowledge-intensive	0.18	0.18	0.25
ICT sector	0.03	0.05	0.21
<i>Female</i>			
Female	0.45	0.42	0.37
Observations	24.6M	3235	3235

Notes: This table shows key characteristics for the population and the final sample, both unweighted and weighted. The population is defined as all persons in Germany who were in regular employment subject to social security contributions on June 30, 2011 without special characteristics, i.e. no civil servants or trainees. In addition, only individuals aged between 16 and 65 or who were not yet over 60 in 2011 are included. The final sample used in our analysis includes people who were working in one of the surveyed firms in 2011 and who otherwise have the same characteristics, with the exception that they are not older than 55 in 2011. In addition, individuals are excluded who have missing information for outcome and treatment variables and have not agreed to the linking of the survey data with administrative IAB data. The weights used are post-stratified design weights as described in Appendix B.2.

B.4 Principal Component Analysis

As already described in Section 2, a Principal Component Analysis (PCA) is a data-driven statistical method that aims to reduce the dimensionality and thus the complexity of (large) data sets while retaining as much of the original variance as possible. This is achieved by recognizing correlations and patterns between the variables and combining them into a small number of new variables. These represent uncorrelated linear combinations of the original numerous correlated variables and are referred to as principal components. They are evaluated in terms of their variance or eigenvalue, i.e. the amount of variation captured in the data, and arranged in descending order. The first principal component therefore has the highest eigenvalue/variance, i.e. it summarizes the most significant characteristics of the data. The order of the principal components is important because the subsequent principal components must be uncorrelated with the previous principal components. This means that the second principal component therefore summarizes the second most variance *under the condition* that it is orthogonal to the first principal component.

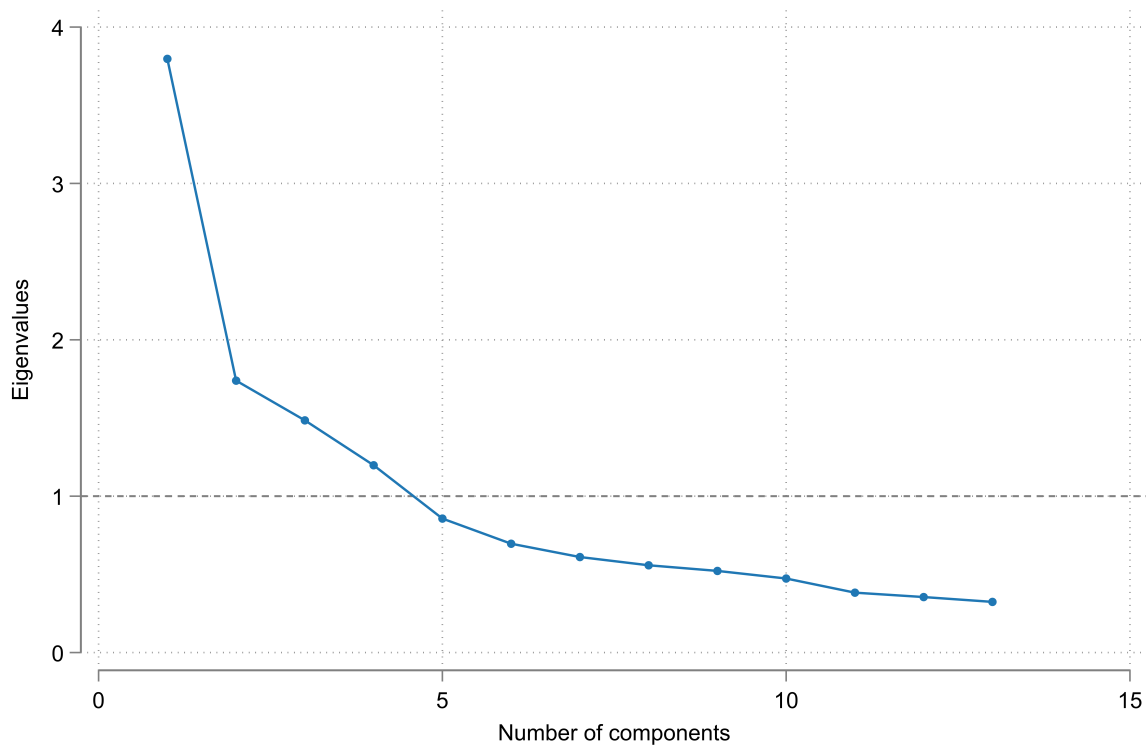
In our case, we want to reduce the number of task dimensions and thus capture groups of employees with similar workplaces and illustrate them more easily. For this purpose, we use 13 variables that describe the workplace of all respondents in 2011 (listed in Table 19).²⁴ Figure 7 shows that these 13 task dimensions can be characterized well with a single component that represents 29.21% of the total variation. As the sharp drop in eigenvalues between components 1 and 2 indicates that the first component captures the largest share of variance in the data and that additional components explain little additional variability, we chose only one component to be retained.

Table 18 shows the loadings, the share of unexplained variation, and the Kaiser-Meyer-Olkin (KMO) measure of our retained single principal component. The KMO measure reflects the suitability of the data for the PCA based on the proportion of common variance among variables and is provided for each variable and overall. It ranges from 0 (bad) to 1 (good); thus, as all variables separately, as well as in sum, have a high value, we can conclude that the data is well-suited. Especially typical office activities like writing, sitting, and the usage of ICT, software, and mathematics have high loadings, i.e., a strong association with the principal component. Workplaces with high loadings are also less repetitive and require less physical exertion, but

²⁴In order to provide the most comprehensive picture possible of workplace characteristics, all 6,923 respondents who were employed in 2011 and had no missings in any of the 13 task dimensions form the basis for the PCA.

are more independent. Based on individuals' workplaces, we assign each worker a *PCA score* (standardized to have mean 0 and a variance of 1) reflecting their degree of office-like tasks performed in 2011.

Figure 7: Scree plot



Notes: This figure shows the eigenvalues of each principal component on the y-axis and the corresponding component number on the x-axis resulting from the Principal Component Analysis (PCA) described in Appendix B.4. The eigenvalues represent the amount of variance captured by each component relative to the original variables. This means that principal components with eigenvalues below one, indicated by the horizontal line, explain less variance than the original variable and should be left out ("Kaiser criterion"). The steep decline at the beginning indicates that the first component explains a substantial amount of variance, while the gradual leveling off suggests that subsequent components contribute less.

Table 18: Loadings of the first principal component

	Loadings	Unexplained	KMO
Writing 2011	0.342	0.555	0.901
Math 2011	0.255	0.754	0.781
Software 2011	0.382	0.447	0.848
Repetitive tasks 2011	-0.218	0.819	0.847
Unpredictable tasks 2011	0.151	0.913	0.691
Physical exertion 2011	-0.379	0.455	0.808
Sitting 2011	0.376	0.464	0.890
Customer contact 2011	0.149	0.915	0.691
ICT 2011: Frequency	0.385	0.438	0.881
Machines 2011: Frequency	-0.256	0.751	0.735
Workintensity 2011	0.090	0.969	0.662
Task-independence 2011	0.222	0.813	0.646
Time-independence 2011	0.154	0.911	0.565
Observations:	6923	Overall KMO:	0.787

Notes: This table presents the loadings, the share of unexplained variation, and the KMO measure for all 13 variables considered in the PCA. Loadings can be interpreted as correlations between the original variables and the component, with a higher absolute value indicating a stronger relationship. The Kaiser-Meyer-Olkin (KMO) measure is provided for each variable, indicating their adequacy for the PCA, with the cumulative KMO measure displayed at the bottom. KMO values range from 0 to 1, with values below 0.5 considered unacceptable, between 0.5 and 0.7 mediocre, between 0.7 and 0.8 good, between 0.8 and 0.9 great, and above 0.9 superb.

Table 19: Workplace Characteristics in 2011

Variable	Description
Work pressure 2011	Rating on 5-point scale of statement "How often do you have to work under sever deadline or performance pressure?"
Self-determination 2011	Rating on 5-point scale of statement "How often can you arrange tasks yourself?"
Self-scheduling 2011	Rating on 5-point scale of statement "How often can you determine your own work pace?"
Repetitiveness 2011	Rating on 5-point scale of statement "How often do you perform repeating identical work processes?"
Unpredictability 2011	Rating on 5-point scale of statement "How often do situations arise that need an individual response?"
Physical exertion 2011	Rating on 5-point scale of statement "How often does your work involve great physical exertion?"
Sitting 2011	Rating on 5-point scale of statement "How often do you work sitting continuously for at least one hour?"
Customer contact 2011	Rating on 5-point scale of statement "How often do you deal with people who are not your colleagues?"
Writing complexity 2011	5 categories: No writing requirements, smaller texts, at least 1-page texts, at least 5-page texts, at least 25-page texts
Math complexity 2011	5 categories: No math, simple calculations, fraction/percent (or similar complex), area calculations (or similar compl.), higher math
Software use 2011	5 categories: No software, specific or standard software, specific and standard software, advanced programming functions, self-programming
ICT frequency 2011	Rating on 5-point scale of statement "How often do you use information and communication technologies?"
Machine frequency 2011	Rating on 5-point scale of statement "How often do you use machines?"

Notes: This table displays the names and a detailed description of all variables considered in the PCA. The 5-point Likert scale ranges from 0 (do not agree/not at all) to 4 (strongly agree/always). For writing and math complexity, as well as software use, respondents were subsequently asked whether their workplace requires them to do certain tasks/work with a certain type of software programs (yes/no). Respondents are then categorized into five categories based on the highest requirement level they report.